

02-12-2025

Regression Components:

$$\begin{aligned}\hat{\beta}_1 &= (\underline{x}_1^T \underline{m}_2 \underline{x}_1)^{-1} \underline{x}_1^T \underline{m}_2 \underline{y} \\ &\quad \because \text{bc } \underline{m} \text{ is sym. + idemp.} \\ &= (\underline{x}_1^T \underline{m}_2^T \underline{m}_2 \underline{x}_1)^{-1} \underbrace{\underline{x}_1^T \underline{m}_2^T}_{\tilde{x}_1^T} \underbrace{\underline{m}_2 \underline{y}}_{\tilde{e}_2} \\ &= (\tilde{x}_1^T \tilde{x}_1)^{-1} \tilde{x}_1^T \tilde{e}_2 \\ &\quad \begin{cases} \text{residuals recovered} \\ \text{residuals of } \underline{x}_1 + \underline{x}_2 \end{cases}\end{aligned}$$

Ch. 4 - Ch. 5 (Lecture Notes 4)

Finite Sample Property of Linear CEF Model:

↳ Unbiasedness, Sampling Variance (finite)

↳ Consistency, Normality (Large)

Assume the following

1. $\underline{y} = \underline{x}^T \underline{\beta} + e$ s.t. $E[e|\underline{x}] = 0$
2. finite moments s.t. $E[y^2] < \infty$ $E\|\underline{x}\|^2 < \infty$
3. positive definite design matrix $E[\underline{x}\underline{x}^T]$

Sometimes

4. homoskedasticity: $E[e^2|\underline{x}] = \sigma^2$

↳ $E[e^2]$ doesn't change given $\underline{x}$

In general, $\hat{\theta}$ of θ : bias $\hat{\theta} = E[\hat{\theta}] - \theta$.
 if bias $(\hat{\theta}) = 0$, $\hat{\theta}$ is unbiased

Exp. of $\hat{\beta}$: $E[\hat{\beta}] = E[E[\hat{\beta} | X]]$

$\hookrightarrow E[\hat{\beta} | X] = E[(X^T X)^{-1} X^T Y | X] \Rightarrow$

$= (X^T X)^{-1} X^T E[Y | X] = (X^T X)^{-1} X^T X \beta$

$\rightarrow$ aside $E[Y | X] = \begin{pmatrix} E[y_1 | X] \\ \vdots \\ E[y_n | X] \end{pmatrix} = \begin{pmatrix} E[y_1 | x_1] \\ \vdots \\ E[y_n | x_n] \end{pmatrix} = \begin{pmatrix} x_1^T \beta \\ \vdots \\ x_n^T \beta \end{pmatrix}$

$= \beta \Rightarrow E[\hat{\beta}] = \beta \therefore \hat{\beta}$ unbiased

$\hookrightarrow \hat{\beta} = (X^T X)^{-1} X^T Y = (X^T X)^{-1} X^T (X \beta + e)$ [aside]

$= (X^T X)^{-1} X^T X \beta + (X^T X)^{-1} X^T e$

$= \beta + (X^T X)^{-1} X^T e$

$E[\hat{\beta} | X] = \beta + \underbrace{(X^T X)^{-1} X^T E[e | X]}_{=0} = \beta$

Sampling Variance of $\hat{\beta}$:

$V_{\hat{\beta}} = \text{var}(\hat{\beta} | X)$: $k \times k$ matrix

$\hookrightarrow$ off-diagonal elements tell covariance

$D := \text{var}(e | X) = E[e e^T | X]$: $n \times n$ matrix

$\hookrightarrow$ diagonal holds $E[e_i^2 | X] = E[e_i^2 | x_i] = \sigma_i^2$

$\rightarrow$ homoskedasticity: σ^2

$\hookrightarrow$ off-diagonal holds $E[e_i e_j | X] = E[e_i | x_i] E[e_j | x_j] = 0 \cdot 0 = 0$

$\rightarrow$ homoskedasticity: $D = \sigma^2 I_n$

$$\text{var}(\underline{Y}|\underline{X}) = \text{var}(\underline{X}\beta + \underline{\varepsilon}|\underline{X}) = \text{var}(\underline{\varepsilon}|\underline{X}) = \underline{D}$$

$$\underline{V}_{\hat{\beta}} = \text{var}(\hat{\beta}|\underline{X}) = \text{var}((\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{Y}|\underline{X})$$

$$= (\underline{X}^T \underline{X})^{-1} \underline{X}^T \text{var}(\underline{Y}|\underline{X}) \underline{X} (\underline{X}^T \underline{X})^{-1}$$

↳ take transpose for conformable

$$= (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{D} \underline{X} (\underline{X}^T \underline{X})^{-1}$$

Under homoskedasticity:

$$\underline{V}_{\hat{\beta}}^0 = \sigma^2 (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{X} (\underline{X}^T \underline{X})^{-1} = \sigma^2 (\underline{X}^T \underline{X})^{-1}$$

Law of Total Variance:

$$\text{var}(Y) = E[\text{var}(Y|X)] + \text{var}(E[Y|X])$$

$$\text{var}(\hat{\beta}) = E[\text{var}(\hat{\beta}|\underline{X})] + \text{var}(E[\hat{\beta}|\underline{X}])$$

$$= E[(\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{D} \underline{X} (\underline{X}^T \underline{X})^{-1}] + \beta$$

$$E[Y] = \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$\text{var}(\bar{Y}) = \frac{\text{var}(Y)}{n}$$

$$\underline{V}_{\hat{\beta}}^0 = \frac{\sigma^2}{n} \left(\frac{1}{n} \underline{X}^T \underline{X} \right)^{-1} = \frac{\sigma^2}{n} \left(\frac{1}{n} \sum_{i=1}^n \underline{x}_i \underline{x}_i^T \right)^{-1}$$